

Person-Hours Estimation: Multiple team members have had experience creating a minesweeper game in the past, so we used aggregated previous experience and used an analogy-based estimation. With a team of 5, we estimated about 20 person hours, as multiple team members in the group had been able to make a similar project by themselves in about 5-7 Hours.

Actual Person-Hours:

Adam Berry	Barrett Brown	Jonathan Gott	Alexander Phibbs	Minh Vu
9/29/2025: Added GUI that improved the experience for starting a game removing the need for command line integration: 4 hours	9/29/2025: Added the AI interaction Skeleton for board solver: 2 hours	9/29/2025: Set up Medium AI sequence for 1-2-1 to be implemented: 1 hour	9/29/2025: Added Easy AI random generation and clicking of tiles: 2 hours	9/29/2025: Started Hard mode AI integration for adding flags: 4 hours
10/2/2025: Worked with Barrett on strange interactions with AI and helper functions to remove bugs: 3 hours	10/2/2025: Worked with Adam on strange interactions with AI and helper functions to remove bugs: 3 hours	10/2/2025: Finished medium AI solver and finalized issues with a timeout: 3 hours	10/2/2025: Helped Minh with Hard mode interactions when playing against AI bots being too fast: 2 hours	10/2/2025: Finished hard mode and helped Alexander with Hard mode interactions when playing against AI bots being too fast: 4 hours
10/3/2025: Added the final stage of the gui to run seamlessly between game start and actual game play even when restarting: 2 hours	10/5/2025: Finished player and bot interaction while doing trial runs to ensure no bugs would take place: 4 hours	10/3/2025: Found a few bugs with players playing against medium solver and removed the problem areas: 3 hours	10/5/2025: Final demonstration of code working with new feature and the documentation completed with new upgrades in mind: 3 hours	